

Effects of the COVID-19 Recession on the US Labor Market: Occupation, Family, and Gender

Presenter: Adeline(Haining) Yuan
Authors: Stefania Albanesi & Jiyeon Kim

September 22th

US recessions usually hit men's employment harder, but the COVID-19 downturn reversed that pattern—women experienced larger employment losses. The authors argue this reflects both demand and supply forces.

Demand side Women are concentrated in service occupations; pandemic mitigation and infection-risk behaviors sharply reduced service demand, so women were disproportionately affected.

Supply side Unlike past recessions when married women often increased labor-force attachment as household “insurance,” closures of schools and childcare during 2020 pushed many parents—especially mothers—out of the labor force.

- **Historical break:** Unlike past U.S. recessions—when men's employment fell more—the COVID-19 downturn saw larger losses for women.
- **Question—demand vs. supply?** Were women hit harder because service-sector demand collapsed, or because school/child-care closures pushed (especially) mothers out of the labor force?
- **Approach** Compare 2007–09 vs. 2020; *classify jobs by work-from-home ability and in-person contact* to identify exposure; then use regressions and gross flows ($E \rightarrow U$ vs. $E \rightarrow N$) to separate demand from supply.

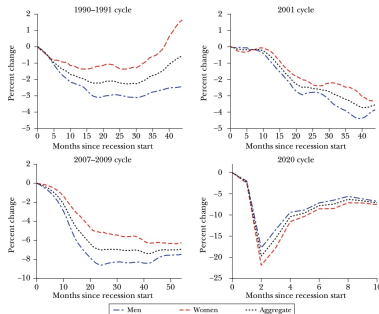
Literature Review

- **Pre-COVID pattern:** Women's labor supply was relatively *countercyclical*; married women were less likely to exit in downturns (added-worker effect; Lundberg 1985; Shore 2010; Blundell et al. 2016).
- **Composition:** Women are concentrated in services, which in typical recessions are less cyclical than goods—helping explain smaller pre-2020 female employment losses.
- **Pandemic shock:** COVID-19 hit high-contact services. Classify jobs by *teleworkability* and *contact intensity* to see which were most exposed (Dingel–Neiman 2020; Mongey–Pilossoph–Weinberg 2020).
- **Family constraints:** School/child-care closures raised caregiving needs; prior work documents a sizable “child penalty” in women's earnings and employment persistence (Kleven–Landaïs–Søgaard 2019; Adda–Dustmann–Stevens 2017).
- **Distributional impacts:** Early job losses were concentrated among lower-wage workers—where women are overrepresented—widening disparities (Cajner et al.; Chetty et al.).

Cycles: Employment-to-Population Across Recessions

Figure 1

Percentage Change in the Employment-to-Population Ratio since the Start of Each Recession for the Four Most Recent Business Cycles



Source: Authors' calculations based on Current Population Survey.

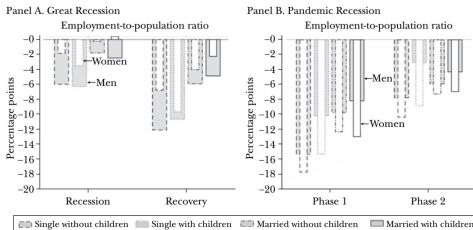
Note: Recession dates based on the National Bureau of Economic Research business cycle dates.

- 1990–1991, 2001, 2007–09: men's employment fell more than women's.
- **2020 reversal:** At the start, women's share working dropped more.
- By 10 months, both improved, but women were still below men relative to pre-COVID.

Gender & Family Status (Great Recession vs. Pandemic)

Figure 2

Change in the Employment-to-Population Ratio Relative to Pre-Recession during the Great Recession and the Pandemic Recession by Gender and Family Status



Source: Author's calculations from CPS.

Note: For the Great Recession, pre-recession corresponds to March–November 2007, Recession to December 2007–June 2009, and Recovery to July 2009–July 2012. For the pandemic recession, pre-recession corresponds to February 2020, Phase 1 to March–May 2020 and Phase 2 to June–November 2020.

- **Great Recession:** losses larger for men across family types.
- **Pandemic:** losses larger for women—largest for *married women with children* in Phase 1; a gap persists in Phase 2.

Occupation Classification (Flexibility $\times$ Contact)

Table 1

Occupation Classification

	Flexible	Inflexible
High-contact	Education, Training, and Library	Healthcare Practitioners and Technical Healthcare Support Food Preparation and Serving Personal Care and Service
Low-contact	Management Business Computer and Mathematical Architecture and Engineering Life, Physical, and Social Science Community and Social Services Legal Arts, Design, Entertainment, Sports, and Media Sales and Related Office and Administrative	Protective Service Building and Grounds Cleaning and Maintenance Farming, Fishing, and Forestry Construction Trades, Extraction Installation, Maintenance, and Repair Production Transportation and Material Moving

Source: Author's calculations based on O*NET.

Note: Occupations are inflexible if their inflexibility score is above the median and flexible otherwise. Occupations are high-contact if the contact intensity score corresponds to a distance of less than 6 feet. Flexibility scores and contact intensity scores are reported in Table 6 in the online Appendix.

- **Two dimensions:** *Flexible* = can be done remotely; *Inflexible* = must be on site. *High contact* = close physical proximity; *Low contact* = farther apart.
- **Measured via O*NET:** Flexibility = avg of 15 task questions (above-median $\Rightarrow$ inflexible); Contact = 1–5 proximity scale (arm's length/near-touching = high contact).
- **Examples:** High-contact & inflexible: healthcare, food prep, personal care. Low-contact & flexible: management, business, computer & math, office/admin.

Occupational Distribution by Gender (Feb 2020 CPS)

Table 2
Occupational Distribution by Gender

Group	Employed women	Employed men	Total employed	Female share
Flexible, High-contact	10	3	6	76
Flexible, Low-contact	53	48	51	50
Inflexible, High-contact	26	9	17	73
Inflexible, Low-contact	11	40	26	19

Source: Author's calculations based on February 2020 CPS.

Note: All values in percentage.

- **Women concentrated in higher-exposure cells:** *Inflexible, high-contact* = 26% of employed women vs. 9% of men (female share **73%**); *Flexible, high-contact* = 10% vs. 3% (female share **76%**).
- **Men concentrated in on-site low-contact work:** *Inflexible, low-contact* = 40% of employed men vs. 11% of women (female share **19%**).
- **Largest cell overall:** *Flexible, low-contact* (about **51%** of all jobs; female share **50%**).
- **Implication:** Because high-contact/on-site jobs were hit hardest by distancing and closures, women's pre-COVID job mix made them more exposed to pandemic job losses.

Regression Framework

Regression Framework

To examine the dynamics of employment over the course of 2020, we estimate the following regression:

$$Y_{i,t} = \alpha + \sum_{\tau=1}^2 \beta_{\tau} \times I(\tau) + \gamma I^i(f) + \delta I^i(m) + \eta I^i(c) + \nu X_t^i + \epsilon_{i,t}.$$

where i indexes an individual and τ is an indicator variable for one of two phases of the pandemic, which are the same as defined earlier for Figure 1, with $\tau = 1$ corresponding to March to May, $\tau = 2$ corresponding to June to November. The variable $I^i(f)$ is a dummy for gender, equal to 1 for female, $I^i(m)$ is a dummy for marital status, equal to 1 for married, and $I^i(c)$ is a dummy for children under the age of 12 present, equal to 1 if they are, and X_t^i include a set of controls for age, educational attainment and, in some specifications, occupation, as categorized in the previous section. Additionally, we include a full set of interactions between the phase effects and the gender, marital status, and presence of children dummies, and the age, education and occupation controls.

Results: Table 3

Table 3
Change in Employment, Unemployment, Nonparticipation by Demographic Groups

Change since February 2020	Employment		Unemployment		Nonparticipation	
	Phase 1	Phase 2	Phase 1	Phase 2	Phase 1	Phase 2
Average without occupation controls	-5.3	-3.8	5.0	3.5	0.3	0.3
Share women	64.3	62.2	63.7	61.1	72.7	78.0
Average with occupation controls	-3.6	-3.1	3.5	3.0	0.1	0.1
Share women	65.9	61.6	65.8	60.0	69.5	121.6

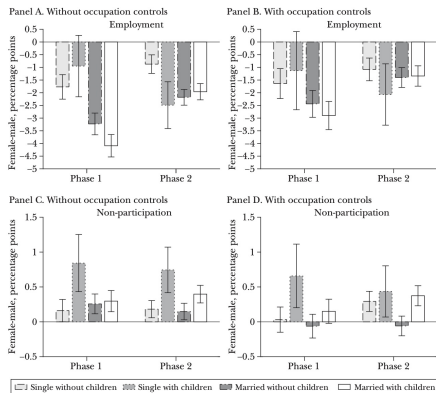
Source: Authors' calculations based on CPS.

Main points

- **Employment declines shrink with occupation controls:** Phase 1: $-5.3\% \rightarrow -3.6\%$ ($\approx$ one-third explained); Phase 2: $-3.8\% \rightarrow -3.1\%$ ($\approx$ one-fifth).
- **Unemployment rise is smaller with occupation controls:** Phase 1: $+5.0$ pp $\rightarrow +3.5$ pp (1.5 pp, $\approx 30\%$ explained); Phase 2: $+3.5$ pp $\rightarrow +3.0$ pp (0.5 pp, $\approx 14\%$).
- **Nonparticipation (exits) is small on average:** $0.3\% \rightarrow 0.1\%$ in both phases.
- **Who bears it:** Women account for ~ 62 – 66% of the employment decline and ~ 61 – 66% of the unemployment rise; for exits, women's share is very high ($\sim 70\%$ in Phase 1 and $> 100\%$ in Phase 2 with controls).

Results: Figure 4 Gender Differences in Employment

Female–Male Difference in Changes in Employment since February 2020,
Estimated with and without Occupation Controls



Source: Author's calculations from Current Population Survey data, using equation in text.

- Women's employment fell more than men's in *every* demographic group.
- Largest gap (Phase 1): married women with children, about **-4 pp** vs. men in the same group.
- Married women w/o children: about **-3 pp** (Phase 1).
- Single parents: **-1 pp** (Phase 1) and **-2.5 pp** (Phase 2).
- Single w/o children: **-2 pp** (Phase 1) and **-1 pp** (Phase 2).

pp = percentage points.

Gross Labor Flows by Demographic Groups (Table 4)

Change in Gross Labor Flows by Demographic Groups

Change since February 2020	EU		UE		EN		UN	
	Phase 1	Phase 2	Phase 1	Phase 2	Phase 1	Phase 2	Phase 1	Phase 2
Average without occupation controls	2.9	1.2	-0.4	-0.6	0.2	0.1	0.1	0.1
Share women	65.1	66.6	57.6	62.1	68.7	68.0	71.0	61.0
Average with occupation controls	1.8	1.1	-0.4	-0.4	0.2	0.1	0.01	0.04
Share women	66.8	58.5	72.4	86.2	55.4	85.0	120.2	76.2

Source: Authors' calculations based on CPS.

- **EU (E→U layoffs):** +2.9 pp (P1), +1.2 pp (P2); with occupation controls +1.8, +1.1 ⇒ about ~1/3 of the Phase-1 surge explained by occupations. Women account for ~65–67% of the increase.
- **UE (U→E job finding):** -0.4, -0.6 (slower hiring), little change with controls. Women's share of the deterioration is high and rising (to ~86% with controls).
- **EN (E→N exit from work):** small but positive (+0.2, +0.1); women's share large (55–85%).
- **UN (U→N discouragement):** tiny ($\approx +0.1$); women's share very high ($> 100\%$ in P1).

Conclusion

- The pandemic recession **reversed** prior cycles: women's employment fell more than men's.
- **Demand channel:** Women's concentration in high-contact, less-teleworkable jobs led to a large **E→U** (layoff) spike; occupation mix explains *about one-third* of the Phase 1 surge.
- **Supply channel: Care constraints** (school/childcare closures) raised **E→N** exits—largest for **mothers**, especially **single mothers**—even after occupation controls.
- **Hiring slowed:** **U→E** declined, with women bearing a larger share.
- **Essential:** Both **demand** and **supply** forces drove 2020 outcomes; **care constraints** are key to the larger losses among mothers.

Limitations & Further Study

What's limited

- Short horizon: evidence through Nov 2020 only (no longer-run persistence).
- No causal ID for school/childcare closures (supply constraints inferred, not estimated).
- Occupation controls (O*NET) are proxies; limited within-occupation detail.
- CPS flows track E/U/N transitions, not hours, schedules, job quality or wages.
- 2020 CPS misclassification risks (e.g., “employed but absent”) may blur U vs N.
- Scope centers on gender $\times$ family; limited intersectional and geographic detail.

Thank you